

# Addition of angular momentum 1

Let  $\mathbf{J}$  be the sum of orbital angular momentum  $\mathbf{L}$  and spin angular momentum  $\mathbf{S}$ .

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

Recall that

$$\begin{aligned} L_x \psi &= -i\hbar \left( y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \psi & S_x \chi &= \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \chi \\ L_y \psi &= -i\hbar \left( z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \psi & S_y \chi &= \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \chi \\ L_z \psi &= -i\hbar \left( x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \psi & S_z \chi &= \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \chi \end{aligned}$$

Let  $\Psi$  be the product of wave function  $\psi$  and vector  $\chi$ .

$$\Psi = \psi \chi$$

Then

$$\mathbf{J}\Psi = \mathbf{L}\Psi + \mathbf{S}\Psi$$

Let  $J^2$  be the magnitude-squared of total angular momentum.

$$J^2 = \mathbf{J} \cdot \mathbf{J} = J_x^2 + J_y^2 + J_z^2$$

Operator  $J^2$  can be decomposed as

$$J^2 = (\mathbf{L} + \mathbf{S}) \cdot (\mathbf{L} + \mathbf{S}) = L^2 + S^2 + 2\mathbf{L} \cdot \mathbf{S}$$

In Eigenmath code

$$\mathbf{L} \cdot \mathbf{S} \Psi = \text{contract}(\mathbf{L}(\mathbf{S}(\Psi))) \quad (1)$$

Verify  $J^2$  commutes with  $L^2$  and  $S^2$ .

$$[J^2, L^2] = [J^2, S^2] = 0$$

Eigenmath script